



Applying Bayes' Theorem The Engine of Inference Transcript

Hey, hi everyone, welcome to the second video of this entire module which is about Bayesian inference. We are going to learn about how do we actually apply Bayesian theorem okay or Bayes theorem. We would learn about the mechanics of Bayes theorem, how can we actually interpret it, what do different terms in Bayesian theorem mean and how do we actually apply in real world scenarios okay.

So the concept here is still the same. We have, we start with certain prior, we update the prior with a new belief given we have got the new data or the belief system is updated given you have new evidence or new data. That's the entire understanding that you need to know about Bayesian theorem.

Other than that it's all about learning to understand how to interpret the mathematical equation but this is very intuitive. If you think of, you initially, humans also believe or they also update the belief in the same way. We all start with certain notion initially and once we receive more information we update that notion with new evidence right.

So given that you have new evidence that you have got, you will update your belief based on that evidence and so entire Bayesian theorem is revolving around the same old concept. This is how your equation actually looks like in the bottom if you think of. I don't know whether someone would have already told you about how does the Bayesian theorem or the entire equation of Bayesian theorem can be interpreted very easily using the concept of prior evidence and so on.

But if you think of on the left hand side we have probability of any new event given or let's say you already have some information about the event. Now you have the probability of theft. Let's say for example the example that we took.

Probability of theft you already have the information about the theft. Now you have got a new information that the car which was used for doing the theft was a red colour Mercedes car. Now you have got the data.

How would this probability could be related with the concepts like likelihood, prior information and the evidence that you have got. The new data that you have got that's what I was telling you about. The new data that you have received is always in the bottom or in the denominator side of it.

Though you also have the likelihood which tells us in actual data how many times you get to see that combination. Let's say for example probability of a theft given it's a Mercedes red car. What is it is exactly equals to probability how many times the red car was actually used to do the theft into probability of theft divided by probability of red Mercedes car in the entire city.



So that's how easily can be written as the same concept of Bayesian or the belief system that you have. Probability of A given B is equals to probability of B given A which is what you are looking into your data. How many times in your data is B given A happening.

Then probability of B which is the prior this the first thing is called as likelihood and at the bottom you have the new evidence that you have got or the data by probability of B. So probability of A given B is equals to probability of B given A this part into the prior term which is probability of A. So you already have some information about probability of A divided by probability of data which is the evidence that you have got. So this is the entire belief system. How do you update or how do you change your belief given you get new evidence every time.

So this is a very simplistic information or introduction to Bayesian. This is what could be the most easiest way of explaining the Bayesian to anyone of you. Let's say for example let me give you another example.

Think of there is a party which is going on and the party you want to find out who has eaten the cake. So someone has eaten the cake. So probability of the cake has been eaten given someone is roaming around the kitchen.

You know that your friend Abhishek or Akriti or whosoever let's say they are roaming around the kitchen. Probability of someone eating the cake given Abhishek roaming around the kitchen is nothing but probability of how many times have you seen Abhishek roaming around the kitchen and the cake was there. Probability of cake and cake theft and then probability of someone your friend roaming around the kitchen.

So how many times actually you have seen or let's say this is the frequentist approach. The frequencies that you have in the data. So you will use that you will use the entire approach to find out what is the possibility or to update the possibility that who actually has taken the cake.

You initially start with certain assumption that let's say individual B who likes cake a lot has taken the cake. That is your probability of theta or the probability of cake taken by someone. Now that is your prior assumption but as soon as you realise that someone else is roaming around the kitchen or was seen roaming around the kitchen maybe in your CCTV camera you would you would find that information.

Now you have found the information that someone was roaming around the kitchen. Initial prior was someone else has taken the cake but now that you have seen someone else roaming around the kitchen your entire belief that who might have taken the cake has changed. So this entire thing is called as updating belief with respect to the new data that or new evidence that you have got.



Now in this entire introduction we have discussed a lot of things. The very first thing was the posterior probability. This is your updated belief as soon as you see a new evidence or new information or new data then you have probability of data given θ which is called as the likelihood how probable is our observed data.

You are talking about the probability of witnessing the data itself and that is why it is called as likelihood. Likelihood and probability they are two different things. In probability you are talking about the outcome.

You are not talking about the data itself. You are talking about the outcome. In likelihood you are talking about the observed data.

What is the probability of observing the data given certain distribution or assuming the certain distribution that is what the likelihood is. So how probable is our observed data given a certain value of the parameter. This actually connects us with the parameter.

Remember we talked about a random parameter and then sketching out the probability distribution around it and so on. Now this is your prior information $P(\theta)$ and then this is your data or new data or new evidence that you have got. Now if you think of this entire Bayesian approach, this entire theorem is a very well scenario which is used in medical diagnosis.

So let us say you have gone and done for a test about a suspicion that you are having a disease but that disease only affects let us say 1% of the population. A test for the disease is 99% accurate. So the test is failing almost 1% of the time.

It detects 99% that you have a particular disease or that particular disease which is also called as the true positive rate. How many true detections are being done by the test. The test has a 5% false positive rate for healthy patients.

So let us say for example for any patient who is not at all having the disease there are 5% cases. So let us say if 100 people go on and use the same test at least 5% or let us say there would be around 5% of the folks who would get wrong test results. That is what it means.

So given that you have gone out and tested positive for a particular test it does not mean that you actually have the disease. So this is a scenario which we can easily explain via Bayesian theorem and we can use the entire approach that we have discussed in the last slides to model the probability of you not having the disease given the test is positive. So probability of A is you not having the disease given the test is positive.

You already have taken the test. The test was positive. Now the question is whether you have the disease or not.



So that is when we will be using Bayesian theorem. So this is the simplest example. Think of it like you have a prior information about the disease from let us say there are only 1% cases of infected population from the disease.

The disease itself is very rare. Now the likelihood is probability of positive given disease. This is what your true positive rate is.

You already have that probability of finding the data given the distribution. So finding positive test given the disease is 99. You already know what is the efficacy of the test.

So you already know the likelihood. Now you are going to calculate the evidence. This is called as total probability.

Probability of positives is nothing but probability of positives given the disease into probability of disease. Probability of positives given no disease into no disease. Because let us say the test can result in positive in both the cases.

One when you actually have the disease. Second when you do not have the disease. So you can be positive in both the cases.

That is why you are actually evaluating the total probability of positive which is comprising of when you have the disease and when you do not have the disease. And then you will simplify this entire thing. You will get a new evidence which is 0.0594. Now you will use the entire Bayesian theorem.

Probability of disease given positive or let us say probability of disease given your test was positive is equals to your likelihood at the top. Then you have the probability of disease which is your prior divided by the evidence. So the evidence that you have just calculated.

It gives you that there are 16.7% chances which is very rare chances. I mean this is still high but 16 chances are very rare chances given that the test is still positive. There are very rare chances that you are having a disease.

So whenever you go out and test and your test becomes positive do not blindly accept the outcome of the test that you are having a disease. There are chances that you might not have the disease as well. So I hope this was this entire sessions the series of sessions the series of videos lectures and everything that we have gone through.

It has given you a lot of information a lot of intuitive understanding and also a background on how these things are developed. There are two different types of approaches that we have discussed one frequentist approach throughout the lecture until this lecture this entire module we discussed about frequentist approach. Now in



this module we are discussing about Bayesian approach and Bayesian approach is mostly driven by conditional probability and there is Bayes theorem.

So we have looked into Bayes theorem we have understood what exactly that is and this entire thing is defined by your prior what you start with then you get a new evidence which is at the denominator. How do you change your posterior or how do you change your belief given you have got the new information this is what the entire theorem says. Thank you everyone.

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